

# Session Objectives

- To recognize bony and soft tissue injuries to the cervical spine
- To appreciate complications related to cervical spine injuries
- To understand management options in spinal injuries
- To recognize bony and soft tissue injuries to the lumbar spine
- To identify urgent and emergent spinal conditions
- To interpret clinical signs and symptoms of spinal pathologies

# History- Spine

- MC chief complaint relating to the spine? Pain
- What are the different kinds of pain? O P Q R S T
  - Localized pain (Tumor/Infection)
  - Mechanical pain (Fracture, Discogenic Disease)
  - Radicular pain (Herniated disc, stenosis)
  - Night pain (tumor)
- What else is important to cover in your history
  - Systemic Symptoms (Fever, Weight Loss, Mental Status Changes) (ROS)
  - Neurologic symptoms (OPQRST, ROS)
  - Trauma history (MIMASH)
  - Occupations (Identify work related injuries, current job and past jobs) (FED TACOS)
  - Recreational activities (FED TACOS)

# Physical Exam

- Inspection
- Gait Analysis
- Palpation of the spine
- Range of Motion
- Neurologic Assessment of trunk and extremities
- Shoulder & Hip evaluation depending on the Chief Complaint
  - Cervical CC → Examine shoulders
  - Lumbar CC → Examine hip and lower extremities
- Special Testing

# Imaging Studies and Lab work

- Plain Radiographs (X-ray; conventional radiography)
  - Usually 1<sup>st</sup> image of choice
  - Can identify acute fractures
- CT Scan
  - More details on bone anatomy
  - Pre-op planning, post-op evaluation, trauma fx
- MRI
  - Better soft tissue resolution
  - Herniated discs, degenerative disease, stenosis, tumor, infection, spinal ligament injury

# Imaging Studies and Lab work

- Bone Scan (Skeletal Scintigraphy)
  - Nuclear medicine test (uses radioactive tracer)
  - Tracer locates areas of Increased metabolic activity
  - Metastatic Disease, Infection, stress fracture
- Blood Tests
  - CBC, CMP, CRP, ESR, SPEP/UPEP

# Cervical Strain / Cervical Sprain

- Etiology
  - Soft tissue activity threshold is surpassed
  - Whiplash Injury (Acceleration-Deceleration Injury)
  - Many cervical muscles do not terminate in tendons but attach directly to the periosteum. Muscles respond to injury by contracting
- Findings
  - Non-focal (distributed throughout), non-radiating neck pain
  - Spasm of paraspinal muscles and trapezius
    - Decreased Range of Motion
    - Upper back and shoulder girdle pain
- Associated with occipital headaches (occipital neuralgia)

# Cervical Spondylosis

- Etiology
  - Chronic disc degeneration and associated facet arthropathy
  - Wear and tear phenomena that occur across the disc-vertebrae interface
  - Usually associated with aging and the inherent loss of water content within the disc itself, becoming stiffer
- Results in **Discogenic neck pain**
  - Insidious onset neck pain, **without FNS**, exacerbated by vertebral motion

Sources: 1. Stockard, Alan, Osteopathic Clinical Joint Exam; American Osteopathic Academy of Sports Medicine; Chapter 1

2. Miller Essential Orthopaedics, 2020. Chapter 108

# Cervical Spondylosis

- Progresses to **radiculopathy**
  - **U/L** neck/shoulder/arm pain, paresthesia, or numbness
  - Exacerbated by mechanical stress on nerve root (motion)
  - **Spurling's Maneuver**
- Severe and chronic cases result in **myelopathy**
  - **Bilateral** weakness (**Upper > Lower**), decreased manual dexterity, **gait disturbances**, sensory defects, **fine motor skills difficulty**, bowel/bladder dysfunction

# Cervical Disc Disease- Imaging and Treatment

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- Radiographs: AP, lateral and oblique views
- CT
  - Shows ossification, foraminal stenosis, and bony cord compress
- MRI
  - Imaging test of choice for cervical, thoracic, and lumbar disc disease
  - Best for spinal cord lesions (syringomyelia, tumors, infection)
- Treatments
  - Conservative
    - NSAIDs (oral steroids if severe), ice/heat, activity modification, soft collar
    - Physical Therapy: Stretching/Traction manipulation/ROM → Isometric exercise / strength → workplace ergonomic mods
  - Surgical
    - Posterior foraminotomy, anterior cervical discectomy & fusion

Source: Miller Essential Orthopaedics, 2020

# Summary- Cervical Spinal Disorders

- Cervical Sprain/Strain: Whiplash, loss of cervical lordosis
- Cervical Spondylosis: Chronic DD, Discogenic pain → Radiculopathy → Myelopathy
- Cervical Disc Disease: Spondylosis + Herniation; Nerve Root BELOW
  - C5 – Biceps
  - C6 – Wrist extensors, brachioradialis
  - C7 – Triceps
  - MRI; Start Conservative treatment
- Rheumatoid Cervical Spondylitis
  - C1-C2 subluxation, transverse ligament of Atlas
  - Skull manipulation → quadriplegia

# Thoracic / Lumbar Spine

- Vertebral Compression Fracture
- Lumbar Herniated Disc
- Discogenic Back Pain
- Spondylolysis
- Spondylolisthesis
- Ankylosing Spondylitis
- Spinal Stenosis
- Cauda Equina

# Lumbar Herniated Disc (Herniated Nucleus Pulposus; HNP)

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- Etiology
  - IV disc degeneration that begins with decreased NP water & proteoglycan content → impairs load-bearing ability
  - Type II collagen (weaker) replaces type I → tears/fissures develop in the AF.
  - Vertebral endplates calcify, impairing nutrient diffusion to the intervertebral disc
  - Progresses to Disc bulging, herniation, facet joint hypertrophy, thickening of ligamentum flavum
- Location
  - Of all symptomatic intervertebral disc herniations, less than 4% occur in the thoracic spine
  - Lumbar herniated NP (HNP); most common at L4-L5, then L5-S1
  - Usually posterolateral (Posterior longitudinal ligament is the weakest)
- If disc impinges nerve roots it causes lumbar radiculopathy
  - I.e. Sciatica if it is one of roots that contributes to the sciatic nerve (L4-S3)

# Lumbar Herniated Disc – Physical Exam

- Observation: changes in symmetry, muscle atrophy, gait analysis
- Palpation
  - Paraspinal muscle spasm/tenderness
  - Sciatic notch tenderness (in radiculopathy)
  - Step-off may indicate associated spondylolisthesis
- ROM testing & special tests

# Lumbar Herniated Disc – Treatment

- Non-operative Treatment (**90% of patients recover in 3 months**)
  - Activity modification, NSAIDS, Ice/Heat
  - PT emphasis on core & lumbar strengthening/stabilization
  - Refractory cases: epidural steroid injections for radiculopathy NOT axial back pain
- Operative Treatment indicated for:
  - Neurologic deficits: spinal cord dysfunction/myelopathy, severe motor deficit, cauda equina syndrome
  - Failed non-operative tx
  - Mitigating psychosocial factors
- Surgical Procedures
  - **HNP** + radiculopathy: **discectomy** (remove disc fragment); >90% success
  - **DDD** with axial low back pain: **fusion**, disc replacement; results inconsistent

# Spondylolysis

- **Overuse injury** of the **lumbar spine** results from increased loading force on the posterior elements (**pars interarticularis** and the pedicles) with repetitive **extension**
- Usually involved activities where the lower back is repetitively hyperextended
  - Gymnastic, Tennis, Volleyball, Dance, Football (lineman)
  - Specific sports-related maneuvers
- Stress injury → Stress fracture

# Spondylolysis - Treatment

- Relative Rest
- Stop the specific activity or sport
- NSAIDs / Tylenol
- RTP is based on resolution of pain and degree of spondylolisthesis (if applicable)
- Physical Therapy
- Consideration for a brace
- Gradual return to play
- In severe cases → surgical stabilization

# Spondylolisthesis

- One vertebral body is slipping relative to another
- 5 Types
  - Dysplastic (abnormal anatomy)
  - Isthmic (caused by a defect in the pars)
    - Most common in young athletes
  - Degenerative (caused by lax facet joints and a degenerative disc that allows the slippage)
    - More common with age
  - Traumatic (high velocity injuries)
  - Pathologic (tumors or osteopenic bone)

# Spondylolisthesis

- Presentation
  - May be asymptomatic
  - May have back pain worsening with extension activities
  - Limited in activities due to back pain, stiffness, and hamstring tightness
- “Step-off” palpable at the deformity
- Grading: % anterior displacement of the fractured vertebrae on the one below
  - 1: < 25%
  - 2: 25-50%
  - 3: 50-75%
  - 4: 75-100%
  - 5: > 100% (Complete displacement)

# Question #1

A 25 year-old man comes to the office due to progressive low back pain and stiffness. The pain has been present for several years but has worsened in recent weeks. The patient has pain relief with physical activity and OTC ibuprofen. Associated symptoms include morning stiffness, that lasts an hour. Since last week he has had some eye redness and blurry vision however he has no rash, urinary symptoms or diarrhea. On exam, he is afebrile. There is limited ROM of the lumbar spine with forward flexion as well as tenderness over the lumbosacral area. What is the most likely diagnosis?

- A. Epidural abscess
- B. Rheumatoid arthritis
- C. Reactive arthritis
- D. Ankylosing Spondylitis
- E. Osteoarthritis of the spine

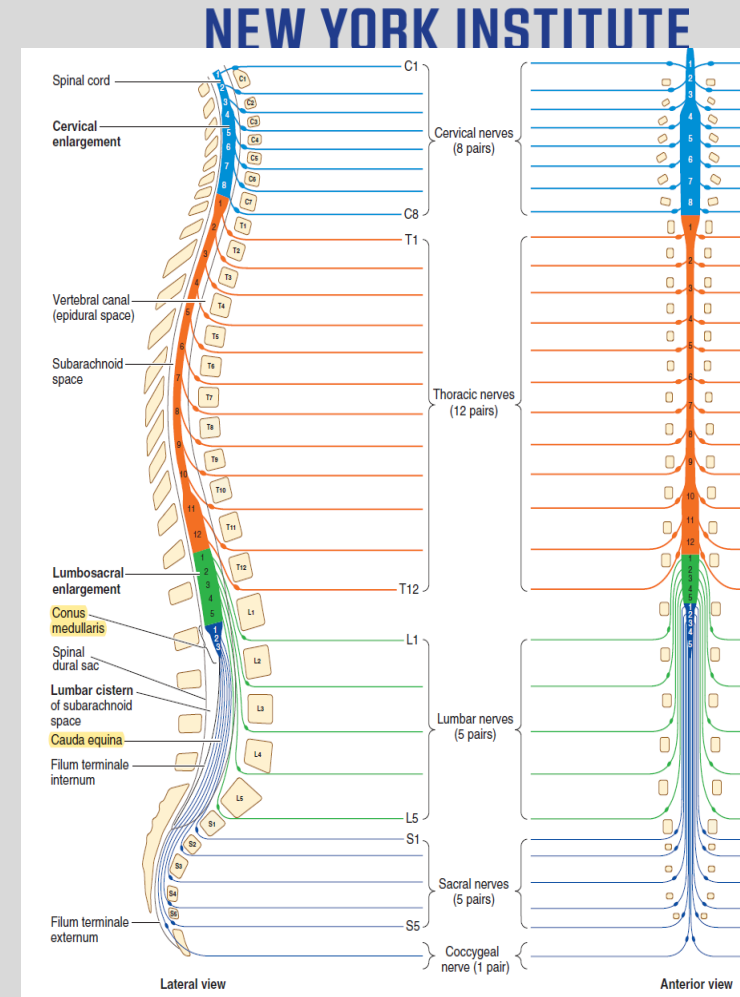
## Question #2

- 67 y/o woman comes to the office for evaluation of low back pain and difficulty walking for 6 months. The pain is worse with prolonged standing and is relieved by sitting. The pain is less severe when walking uphill than walking downhill. On examination, there is no spinal tenderness and a negative SLR. Reflexes are 2+ in both legs and distal pulses are intact. What is the most likely diagnosis?
  - A. Herniated Lumbar Disc with Radiculopathy
  - B. Cauda Equina Syndrome
  - C. Peripheral Vascular Disease
  - D. Spinal stenosis
  - E. Vertebral Compression Fracture

# Cauda Equina Syndrome

- Spinal cord ends at the **conus medullaris (L1)**
- Below this level spinal canal filled with L2-S4 nerve roots
  - Free floating nerve roots look like a horse's tail & thus called cauda equina
- Compression leads to **flaccid paralysis** (different from spastic paralysis seen with myelopathy)
  - The cauda equina are **PERIPHERAL** nerves
- Etiologies
  - Massive central lumbar disc herniation
  - Fracture, Trauma
  - Epidural abscess or hematoma
  - Neoplasm
  - Infection

Sources: Miller Essential Orthopaedics, 2020. Chapter 114.  
Moore Clinically Oriented Anatomy, 7<sup>th</sup> ed



# Cauda Equina Syndrome

- Presentation
  - Pain lower back/ buttocks
    - Usually unilateral with severe radicular pain
  - Saddle anesthesia
  - Loss of anal sphincter tone & wink reflex
  - Bowel incontinence
  - Urinary retention → Overflow urinary incontinence
  - Flaccid paralysis & decreased LE reflexes
- Diagnostic Imaging
  - MRI (test of choice)
  - CT Myelography (if unable to do MRI)
- Surgical Emergency!!!
  - Decompression of cauda equina within 48hrs of symptoms onset

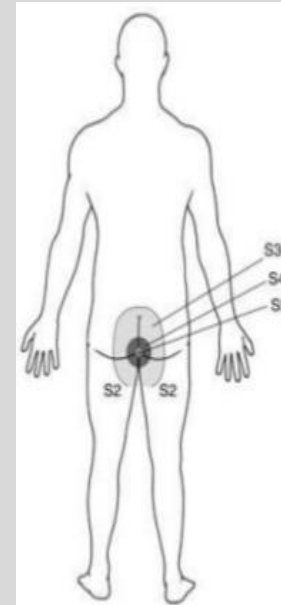


Illustration of saddle anesthesia;  
- The S5, S4, and S3 nerves provide sensory innervation to the rectum, perineum, and inner thigh.

Source:  
Journal of the American Academy of Orthopaedics Surgeons,  
<http://www.jaaos.org>

# Summary- Thoracic/Lumbar Spinal Disorders

- **Vertebral Compression Fracture:**
  - Osteoporosis, SP point tenderness, thoracic spine, bisphosphonates
- **Lumbar Herniated Disc: NP/AF degen → posterolateral herniation**
  - L4 – Quad weakness, patellar reflex
  - L5 – dorsiflexion, heel walking
  - S1 – plantarflexion, toe walking, “achilleS-1 weakness/reflex”
- **Discogenic Back Pain:**
  - DDD w/o herniation or radiculopathy
- **Spondylolysis:**
  - hyperextension stress fx, “scotty dog”-sign, gymnast
- **Spondylolisthesis:**
  - ant. Displacement, “step-off” sign, L4 on L5
- **Ankylosing Spondylitis:**
  - 20-30s, HLA-B27, chest expansion/pulm compromise, uveitis, aortic regurg, sacroiliitis, bamboo spine
- **Spinal Stenosis:**
  - ANY degenerative narrowing, pain w/ standing, “shopping cart”-sign
- **Cauda Equina**
  - Massive peripheral nerve damage past L1, flaccid, bladder, bowel, saddle anesthesia
  - EMERGENT SURGICAL DECOMPRESSION!!!

# Lecture Summary

- Be familiar with the history, physical, and basic imaging needed for evaluation of the spine
- Cervical Spine
  - Understand the mechanism and imaging of cervical sprain/strain
  - Understand the mechanism and presentation of cervical spondylosis & rheumatoid Spondylitis
  - Know the mechanism, pertinent exam findings, and management of cervical disc disease
- Thoracic/Lumbar Spine
  - Know the mechanism, pertinent exam findings, and management of vertebral compression fractures and herniated disc
  - Recognize discogenic back pain
  - Be able to differentiate between spondylolysis and spondylolisthesis
  - Understand the presentation, exam findings, labs and imaging of ankylosing spondylitis
  - Understand the mechanism and presentation of spinal stenosis
  - Recognize the presentation of cauda equina syndrome and know the management

# Question #3

A 70 y/o female is brought into the ED by EMS for difficulty breathing. She has been battling a respiratory viral infection for the last few days that has since worsened. On physical exam, you notice B/L swelling of the MCP and PIP joints of the hand. Her daughter is present and tells you that her mother has a hx of atrial fibrillation, stroke, and autoimmune disease. Shortly after arriving to the hospital, the patient begins to decompensate and you worry that she will soon be unable to breath for herself. You call the rapid response team and proceed with intubation. The following day, while in the ICU, the patient regains consciousness however she is now unable to move her arms or legs. What is the most likely cause of the patient's quadriplegia?

- A. Ruptured berry aneurysm
- B. Embolic stroke
- C. Spinal Cord compression between C1 and C2
- D. Cervical disc herniation
- E. Infectious CNS Abscess

# Question #4

Walter White is a 52 year old high school chemistry teacher who moonlights in fast food restaurants for extra money. He comes into your office c/o right arm pain for last few months. He states that he has been having trouble lifting heavy packages and using the cash register with that arm. On physical exam you notice weakness in right wrist extension as well as sensory defects over the lateral right forearm. You are able to reproduce Walter's pain by extending and rotating his neck followed by the addition of a compressive force on his head.

What would you diagnose Walter with?

- A. C4/C5 disc disease with radiculopathy
- B. C5/C6 disc disease with radiculopathy
- C. Cervical spondylosis with myelopathy
- D. C6/C7 disc disease with radiculopathy

# Question #5

Walter White is a 52 year old high school chemistry teacher who moonlights in fast food restaurants for extra money. He comes into your office c/o right arm pain for last few months. He states that he has been having trouble lifting heavy packages and using the cash register with that arm. On physical exam you notice weakness in right wrist extension as well as sensory defects over the lateral right forearm. You are able to reproduce Walter's pain by extending and rotating his neck followed by the addition of a compressive force on his head.

What would you expect to find on reflex testing?

- A. Bicep hyporeflexia
- B. Bicep hyperreflexia
- C. Brachioradialis hyporeflexia
- D. Brachioradialis hyperreflexia
- E. Tricep hyporeflexia
- F. Tricep hyperreflexia